

Study	Non-neural arms	Online	Density
Aksu et al. [2024]	Naive, SNaive, AutoARIMA, AutoETS, AutoTheta	no	no
Ansari et al. [2024]	the above, plus the SCUM ensemble	no	no
Ansari et al. [2025]	the above; task-specific deep models dropped	no	no
Das et al. [2024]	ETS, ARIMA, CatBoost	no	no
Auer et al. [2025]	AutoARIMA, AutoTheta, SNaive	no	no
Woo et al. [2024]	AutoARIMA, SNaive	no	no
Shchur et al. [2025]	Naive, SNaive, Drift, the three above, ensemble	no	no

Table 1: Non-neural arms in the studies that rank time-series foundation models. *Online* means the method carries state between forecasts rather than refitting. *Density* means it issues a predictive density rather than a point or a quantile vector. No arm in any study is either.

A Stronger Univariate Benchmark for Foundational Time-Series Model Evaluation

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Abstract. A foundation model forecasts from a fixed window, with weights frozen at pretraining. An online forecaster carries state and revises it at every observation. Benchmarks for the first kind contain no example of the second: of the 132 entries on the GIFT-Eval leaderboard, five are not neural, and all five refit from scratch on each window. We propose **laplace**, a zero-dependency online distributional forecaster with no pretraining and no tunable parameters, as the missing reference. Scored one step ahead on held-out log density over an 18,815-series FRED panel, it beats TimesFM 3.0 on the median series in every stratum by roughly half a nat, and a post-hoc portfolio that sees only the model’s own issued density recovers most of the gap without retraining. The reference costs a `pip install` to include, and a **Rust** core with a bit-exact parity suite makes its numbers recompute on other hardware.

Keywords: time series, foundation models, benchmarking, probabilistic forecasting, proper scoring rules, online learning.

1 Introduction

A foundation model forecasts from a fixed window of recent history, with weights frozen at pretraining. An online forecaster carries state, and revises it with every observation it sees. Benchmarks for the first kind contain no example of the second.

Table 1 lists the non-neural arms of the studies that rank these models. The set barely moves between them: naive, seasonal naive, and the three **statsforecast** calls, sometimes with the **SCUM** ensemble of Petropoulos and Svetunkov [2020]. Each study also carries deep models, but those are trained per dataset and serve as competitors rather than as references.

None of these arms is online. The **statsforecast** predict path is handed an array, fits, forecasts, and keeps nothing, so every window starts over. In Ansari et al. [2024] and in

the Monash tables of [Das et al. \[2024\]](#) there is a single forecast origin per series, so the question does not arise.

None of them issues a density either. Their quantiles come from a Gaussian residual-variance interval, which the harness reads off level by level, so the predictive is symmetric by construction with no skew and no tail. The ensemble is worse: its quantiles are a per-level median across members, which is not a distribution at all.

NPTS would fill the gap. It is training-free, fully distributional, and ships inside **GluonTS**, which these harnesses already import. It appears in none of the studies in Table 1.

This paper supplies such an arm and reports what it does to the comparison. The contribution is deliberately small: a reference, a protocol, and four tables.

2 **laplace**: an online distributional forecaster

laplace imports nothing, requires no GPU, performs no fitting step, and exposes no tunable hyperparameters. One parameter-free call returns a predictive density.

The predictive object is a weighted mixture of Gaussians. That choice buys three things a reference arm needs: the mixture is closed under affine maps and under mixing, it admits a closed-form CRPS, and it approximates heavy tails as a scale mixture. A density can therefore be pushed through an invertible map and scored against a proper rule [\[Gneiting and Raftery, 2007\]](#) without sampling.

A *transform* is a stateful causal map that consumes one observation and emits one transformed value, paired with an inverse that rebuilds predictive distributions for future observations from the transformed coordinate, given the state it has accumulated. Differencing, standardisation, online autoregression, **GARCH**-style scaling, seasonal differencing and the Yeo–Johnson coordinates are all transforms. *Conjugation* composes one with an inner forecaster: the observation is transformed, the inner model issues a density in the new coordinate, and the inverse carries that density back. Multi-step uncertainty accumulates through the inverse rather than by simulation.

laplace is one fixed composition of these parts, with no knob exposed. Candidate models enter it only as mean forecasters, weighted online by predictive likelihood, so that a shaped leaf governs the tail rather than the dispersion of candidate means. A lattice projection places atoms on the exact values a grid-quoted series revisits, and vanishes identically on continuous data. Beyond one step the predictive mixes over decimated clocks, since the fine clock wins short horizons and a coarse one wins long. A conditional tail fit splices the extreme quantiles, which a bulk fit leaves too thin.

[Cotton \[2026\]](#) derives all of this. We restate only what a reader needs to judge the comparison.

Two properties matter for a reference arm, and neither is accuracy. The first is cost. Adding **laplace** to a leaderboard is a `pip install` and one function call, with no weights to download, no device to allocate, and no context length to choose. The pure-Python implementation, 280 KB of source across 28 files, is the reference, and a Rust core steps it at roughly 14 microseconds per observation, twenty times the Python rate. A reference that is slow over a large panel does not get included.

The second is reproducibility across machines. The Rust core uses portable mathematics (**libm**) and produces identical bits on x86 and ARM, and a shared parity suite checks roughly 100,000 probe values to 10^{-6} per implementation. A number one person publishes therefore recomputes on another person’s hardware, which is the property a denominator needs and a pretrained checkpoint behind an inference stack does not have.

Arm	LL win (raw / family)	CRPS win (raw / family)	N
AutoARIMA (statsforecast)	87/87 %	58/63 %	1688
AutoETS (statsforecast)	95/94 %	77/78 %	1688
Theta (R forecast)	92/95 %	74/79 %	1917
auto.arima (R forecast)	84/85 %	51/60 %	1917

Table 2: Share of series on which **laplace** scores better, over 5,397 continuous non-price FRED series in 1,213 correlated families. The family figure averages within a family first, so that a cluster of related series counts once. N is the subset each arm covers under its compute budget.

A denominator sets what a skill score means, and a weak one flatters everything divided by it. That is the argument for this particular arm.

Sections 4 and 5 measure how strong it is. **laplace** takes the per-series log-likelihood against every classical arm in Table 1, and against TimesFM 3.0 in every stratum tested. Cotton [2026] reports the same ordering against neural and foundation-model baselines across the non-price FRED universe; **GARCH- t** on asset prices is the one arm that escapes, and that split is reported rather than averaged away.

3 Protocol

Every arm sees the same series, the same windows and the same scoring rule. The context is 128 observations, the test window rolls one step ahead, and no arm is fitted to the evaluation data. The score is held-out log predictive density, differenced per series so that each series’ intrinsic difficulty cancels.

We report the median of that difference and the per-series win, draw and loss counts from a Diebold–Mariano test with HAC standard errors and a draw band, following Diebold and Mariano [1995]. We do not report a pooled mean. Averaging a log density across heterogeneous series estimates nothing, because a few near-constant series dominate the sum; Koning et al. [2005] made the corresponding argument about the M3 competition, and Hewamalage et al. [2023] prescribe per-series testing.

The context is fixed rather than growing. Several foundation models degrade beyond the window they were pretrained on, and most of them silently truncate over-long input rather than raising, so feeding four thousand points to a model configured for two thousand scores a two-thousand-context model without saying so. A protocol that lets context grow needs a declared per-model cap, set at the documented architectural maximum, and recorded per arm.

Recording it matters for a second reason. The Woo et al. [2024] benchmark numbers come from a per-dataset search over context lengths from one thousand to five thousand, jointly with patch size, which the arms it is compared against do not receive.

4 Against the classic arms

The arms of Table 1 are the denominator every published skill score divides by, so their strength sets what a skill score means. Table 2 scores **laplace** against them on the non-price FRED universe.

On log-likelihood the margin is wide: the closest arm loses on 84 % of series and the widest on 95 %. On CRPS the arms do better, which is the expected pattern, since CRPS

Stratum	N	Win / draw / loss	Median ΔLL	Loss rate
Daily, economic	2,179	66/923/1,190	−0.854	54.6 %
Daily, price/returns	7,633	15/4,593/3,025	−0.616	39.6 %
Weekly, economic	3,013	517/1,481/1,015	−0.520	33.7 %
Monthly, economic	5,576	193/3,592/1,791	−0.647	32.1 %
M4-hourly, seasonal	414	8/241/165	−0.543	39.9 %

Table 3: TimesFM 3.0 against `laplace`, one step ahead on held-out log density, 128-length context. ΔLL is the per-series difference, model minus reference, reported as a median. The loss rate is the share of series on which the model loses.

Stratum	N	Win / draw / loss	Median ΔLL	Loss rate
Daily, economic	2,179	76/1,383/720	−0.094	33.0 %
Daily, price/returns	7,633	152/7,136/345	−0.008	4.5 %
Weekly, economic	3,011	467/2,045/499	−0.049	16.6 %
Monthly, economic	5,195	350/4,480/365	−0.056	7.0 %
M4-hourly, seasonal	414	58/352/4	−0.007	1.0 %

Table 4: The same model wrapped in a never-worse portfolio with the reference. The loss rate falls by as much as thirty-nine points; the median stays negative in every stratum.

is forgiving of a mis-shaped tail that a log score punishes.

This is the sense in which the incumbent denominators are soft. A skill score computed against them credits a model for clearing a bar that a dependency-free online method clears by a wide margin on the same data.

5 Against TimesFM 3.0

Table 3 scores TimesFM 3.0 [Das et al., 2024] zero-shot against the reference on an 18,815-series FRED panel. Negative favours the reference.

The model loses on the majority of series in every stratum, by a median of roughly half a nat. Half a nat is a large gap in log-likelihood terms.

Table 4 repeats the comparison with the model wrapped in a post-hoc portfolio that combines its own issued predictive with the reference. The wrap requires no retraining and no access to model internals; it sees only the density the model published.

Two readings follow. The wrapped model is safe to deploy, since its loss rate collapses. It is still not competitive, since the median remains negative everywhere.

The second reading is the more useful one. If a post-hoc combination that sees only the published density recovers most of the gap, then at this horizon the pretraining is contributing less than the raw comparison suggests, and what it does contribute survives a wrapper.

6 The reconstruction moves the answer more than the model does

A quantile-only model does not emit a density, and a density must exist before a log score does. Something must build one from the quantiles, and that something is a modelling choice independent of the model being scored.

Fix TimesFM 3.0 and fix 3,200 held-out test points, then vary only the reconstruction across four methods. The mean log density moves from -0.373 to $+2.104$. The mean per-point spread is 3.13 nats, which exceeds the entire measured gap between the model and the reference in every stratum of Table 3. Two methods already in independent use elsewhere differ by 0.62 nats.

The choice therefore belongs in the protocol, stated rather than assumed. It is not peculiar to this study: the Aksu et al. [2024] submission interface requires a mean and nine deciles, so a model emitting a genuine density must reduce it before scoring and a model emitting samples must convert.

The conversion has teeth. That interface’s interval score asks for the 2.5th and 97.5th percentiles, and a sample forecast returns a nearest order statistic without interpolation, so at the twenty samples the reference **Chronos** notebook uses, those two quantiles are the minimum and the maximum of twenty draws.

7 Horizon, grid, and statefulness

The results above do not contradict any published leaderboard, because no published leaderboard occupies this regime. Three settings separate them.

The horizon never reaches one step. The shortest prediction length in Aksu et al. [2024] is six, and Shchur et al. [2025] use domain horizons such as thirty and one hundred sixty-eight.

The scoring rule is a nine-point grid. What these suites report as CRPS is a mean of weighted quantile losses at the deciles, so nothing outside the tenth and ninetieth percentile enters any published score. Ansari et al. [2025] call that grid a defect and move to twenty-one levels including the first and ninety-ninth.

And no baseline carries state, which is the hole of Section 1. Each setting is defensible for the task those suites set themselves. Together they define a regime, and an online distributional method cannot compete inside it or be seen to.

8 Scope

Everything measured here is univariate and one step ahead. A single pretrained model amortized across many related series can exploit cross-series and panel structure that no univariate method reaches by construction, and that is where pretraining should pay for itself most directly. This paper does not make that comparison, and it is the more decisive one.

The protocol is assembled from standard parts. Proper scoring rules are the norm in this literature; prequential test-then-train evaluation is textbook; paired per-series testing was urged on the field by Koning et al. [2005] and is already the headline aggregation in Shchur et al. [2025]; normalizing by a cheap reference is universal. The assembly is what is new.

Leakage is controlled in this literature by dataset name rather than by time. Ansari et al. [2024] state the temporal standard, that evaluation data should begin after the last pretraining observation, and then decline it as a minimal risk. Meyer et al. [2025] and Oreshkin et al. [2026] report that a corpus used in TimesFM pretraining reappears in its Monash evaluation; their pretraining table lists Traffic at 862 series and 15,122,928 observations, and the Monash archive file [Godaheva et al., 2021] holds 862 series of exactly 17,544 steps.

A calibration framing would have to engage Adler et al. [2026], who find these models better calibrated than automated ARIMA and N-BEATS on standard datasets. Nothing here contradicts that. Calibration on a decile grid and sharpness of a full density are different measurements.

Finally, the panel is FRED, and FRED is the universe these models’ pretraining corpora almost entirely omit: the corpus of Woo et al. [2024] is 0.09 % economic by observation, and the entire economics domain of Aksu et al. [2024] is the M4 collection. Whether the gap survives on the energy and transport data these models were built for is an open question, and Wang et al. [2026] suggest it will not.

9 Reproduction

The paper directory holds four files. `constants.py` derives every number in Sections 5 and 6 from a version-controlled store. `verify_paper.py` re-derives each one, compares it against the tables above, and exits non-zero on drift. `REPRODUCE.md` gives three tiers: the figures check in seconds from committed files, the summaries rebuild from per-step records in minutes, and the records regenerate from raw data in days.

The data vintage is published as a release asset in the repository `skaters-benchmarking-data`. FRED revises series, so a fresh pull returns different vintages and will not reproduce these numbers. The archive carries 210,574 public-domain series; commercial index series are excluded and listed, which costs the price stratum and nothing else.

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